

CML — Diagnosis & Lab Workup



1. Cytogenetics mandatory

WHAT YOU SEE

CYTOGENETICS panel on right wall, MANDATORY, karyotype of chr 9 and 22

WHAT IT ENCODES

Bone-marrow karyotype (conventional cytogenetics on ≥ 20 metaphases) is mandatory at diagnosis: it confirms the Ph chromosome AND surveys the whole genome for additional clonal abnormalities (ACAs) that FISH and PCR cannot see. It also establishes the baseline against which cytogenetic response is later measured.

BOARD TRAP

Karyotype — not FISH or PCR — is the only test that detects additional cytogenetic abnormalities (+8, i(17q), double Ph, 3q26.2). These ACAs upstage to accelerated phase and change risk, so you cannot skip the marrow karyotype at diagnosis.

2. Ph t(9;22)

WHAT YOU SEE

Chromosomes 9 and 22 shown in karyotype panel

WHAT IT ENCODES

Demonstrating the Philadelphia chromosome — the derivative chr22 from t(9;22)(q34;q11) — establishes the diagnosis of CML in the right clinical/lab context. Found in nearly all metaphases at diagnosis; molecular confirmation of BCR-ABL1 is still required.

BOARD TRAP

Diagnosis needs BCR-ABL1 confirmation: a 'leukemoid reaction' (infection/sepsis) can mimic the CBC but is Ph-negative with a HIGH LAP score.

3. ACAs (additional abnormalities)

WHAT YOU SEE

ACAs label under cytogenetics; FISH 'cannot detect ACAs'

WHAT IT ENCODES

Additional cytogenetic abnormalities are clonal changes beyond the Ph chromosome (trisomy 8, i(17q), a second Ph, 3q26.2/EVI1). They reflect clonal evolution, predict inferior outcomes, and their presence is an accelerated-phase criterion. Only full karyotype detects them.

BOARD TRAP

ACAs are prognostically important but invisible to FISH/PCR — this is the board's favorite reason that karyotype remains mandatory despite molecular testing.

4. FISH

WHAT YOU SEE

FISH panel showing paired fluorescent signals

WHAT IT ENCODES

FISH uses fluorescent BCR and ABL1 probes to detect the fusion signal rapidly (can be done on blood, not just marrow) and confirms cryptic or variant translocations missed by karyotype. Useful for quick diagnostic confirmation.

BOARD TRAP

FISH confirms the BCR-ABL1 fusion but CANNOT detect additional cytogenetic abnormalities or measure transcript level on the International Scale — it does not replace karyotype (for ACAs) or PCR (for monitoring).

5. Quantitative PCR

WHAT YOU SEE

PCR panel with amplification trace

WHAT IT ENCODES

Quantitative RT-PCR measures BCR-ABL1 transcript level on the International Scale (IS) and is the most sensitive test (detects ~ 1 leukemic cell in 100,000). It is the gold standard for monitoring response and minimal residual disease — the baseline is set at diagnosis ($\sim 100\%$ IS) and tracked at 3/6/12 months.

BOARD TRAP

PCR is the most sensitive test and the one used for serial monitoring/MRD and milestone assessment — not karyotype. Standard primers can miss atypical transcripts (false-negative).

6. Leukocytosis

WHAT YOU SEE

Dense crowd of stick figures on the left wall (pre-TKI panel) — the packed crowd IS the massive leukocytosis with a full left-shifted myeloid spectrum

WHAT IT ENCODES

The classic CBC shows marked leukocytosis (often $> 100,000$) with a left-shifted myeloid series displaying the FULL maturation spectrum — myelocytes, metamyelocytes, bands, neutrophils — with a characteristic 'myelocyte bulge.' Often with thrombocytosis and mild anemia. Many patients are asymptomatic at diagnosis (incidental high WBC) or present with fatigue, splenomegaly, or early satiety.

BOARD TRAP

Unlike acute leukemia, CML shows the complete maturation spectrum with relatively FEW blasts in chronic phase — a 'leukoerythroblastic' full-range left shift, not a blast-dominant picture.

7. Low LAP score

WHAT YOU SEE

Faded/dim neutrophil figures in the crowd — their low glow stands for the LOW leukocyte alkaline phosphatase (LAP) of malignant CML neutrophils

WHAT IT ENCODES

Leukocyte alkaline phosphatase (LAP, also called neutrophil ALP / NAP score) is LOW or absent in CML because the malignant neutrophils are functionally abnormal. This is a cheap discriminator from a reactive leukemoid response.

BOARD TRAP

LAP is LOW in CML but HIGH in a leukemoid reaction, infection, and other myeloproliferative neoplasms (e.g. polycythemia vera). High WBC + LOW LAP $\rightarrow$ think CML; high WBC + HIGH LAP $\rightarrow$ reactive.

8. Basophilia

WHAT YOU SEE

Basophil/eosinophil markers tagged on the accelerated band of the central pyramid — basophilia as a CML hallmark that climbs toward the $\geq 20\%$ accelerated cutoff

WHAT IT ENCODES

Peripheral basophilia (often with eosinophilia) is a near-universal CML hallmark and a useful diagnostic clue. The degree of basophilia rises with progression — basophils $\geq 20\%$ is an accelerated-phase criterion.

BOARD TRAP

Persistent unexplained basophilia should prompt a workup for CML. Basophils $\geq 20\%$ specifically upstages to accelerated phase.

9. Chronic $<5\%$ blasts

WHAT YOU SEE

CHRONIC platform $<5\%$ blasts

WHAT IT ENCODES

At diagnosis the vast majority ($\sim 90\%$) of patients are in chronic phase with $<10\%$ blasts in blood and marrow (this drawing shows $<5\%$). Chronic phase is well controlled by TKIs with near-normal life expectancy.

10. Blast % defines phase

WHAT YOU SEE

BLASTIC $\geq 30\%$ explosion at top

WHAT IT ENCODES

Blast percentage in blood/marrow is THE key parameter staging the disease: chronic $<10\%$, accelerated 10-19%, blast phase $\geq 20\%$ (WHO). Always quantify blasts on the diagnostic marrow.

BOARD TRAP

Memorize the cutoffs: chronic $<10\%$, accelerated 10-19%, blast crisis $\geq 20\%$ blasts. Some older criteria use $\geq 30\%$ for blast phase — know both.

11. Baseline BCR-ABL 100%

WHAT YOU SEE

100% diagnosis anchor on bottom bar

WHAT IT ENCODES

The baseline BCR-ABL1 transcript level ($\sim 100\%$ on the International Scale) is established at diagnosis so that every subsequent log reduction can be measured against it. All response milestones ($\leq 10\%$ at 3mo, $\leq 1\%$ /CCyR at 12mo, MMR $\leq 0.1\%$) reference this baseline.

12. MR2/CCyR $\leq 1\%$

WHAT YOU SEE

MR2 CCyR $\leq 1\%$ milestone

WHAT IT ENCODES

Complete cytogenetic response (CCyR) corresponds to BCR-ABL1 $\leq 1\%$ IS (a 2-log reduction, MR2) — an important early treatment goal, ideally by 12 months. The diagnostic baseline and these milestone bands frame the whole monitoring strategy.
